SQL Window Functions Cheat Sheet

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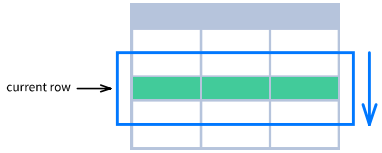
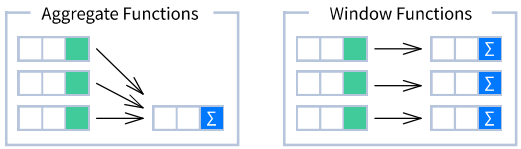
WINDOW FUNCTIONS

Window functions compute their result based on a sliding window frame, a set of rows that are somehow related to the current row.



AGGREGATE FUNCTIONS VS. WINDOW FUNCTIONS

Unlike aggregate functions, window functions do not collapse rows.



SYNTAX

```
SELECT city, month,
  SUM(sold) OVER (
    PARTITION BY city
    ORDER BY month
    RANGE UNBOUNDED PRECEDING) total
FROM sales;
```

```
SELECT <column_1>, <column_2>,
  <window_function> OVER (
    PARTITION BY <...>
    ORDER BY <...>
    <window_frame>) <window_column_alias>
FROM <table_name>;
```

NAMED WINDOW DEFINITION

```
SELECT country, city,
  RANK() OVER country_sold_avg
FROM sales
WHERE month BETWEEN 1 AND 6
GROUP BY country, city
HAVING sum(sold) > 10000
WINDOW country_sold_avg AS (
  PARTITION BY country
  ORDER BY avg(sold) DESC)
ORDER BY country, city;
```

```
SELECT <column_1>, <column_2>,
  <window_function>() OVER <window_name>
FROM <table_name>
WHERE <...>
GROUP BY <...>
HAVING <...>
WINDOW <window_name> AS (
  PARTITION BY <...>
  ORDER BY <...>
  <window_frame>)
ORDER BY <...>;
```

`PARTITION BY`, `ORDER BY`, and window frame definition are all optional.

LOGICAL ORDER OF OPERATIONS IN SQL

1. FROM, JOIN
2. WHERE
3. GROUP BY
4. aggregate functions
5. HAVING
6. **window functions**
7. SELECT
8. DISTINCT
9. UNION/INTERSECT/EXCEPT
10. ORDER BY
11. OFFSET
12. LIMIT/FETCH/TOP

You can use window functions in `SELECT` and `ORDER BY`. However, you can't put window functions anywhere in the `FROM`, `WHERE`, `GROUP BY`, or `HAVING` clauses.

PARTITION BY

divides rows into multiple groups, called **partitions**, to which the window function is applied.

month	city	sold
1	Rome	200
2	Paris	500
1	London	100
1	Paris	300
2	Rome	300
2	London	400
3	Rome	400

`PARTITION BY city`

month	city	sold	sum
1	Paris	300	800
2	Paris	500	800
1	Rome	200	900
2	Rome	300	900
3	Rome	400	900
1	London	100	500
2	London	400	500

Default Partition: With no `PARTITION BY` clause, the entire result set is the partition.

ORDER BY

`ORDER BY` specifies the order of rows in each partition to which the window function is applied.

sold	city	month
200	Rome	1
500	Paris	2
100	London	1
300	Paris	1
300	Rome	2
400	London	2
400	Rome	3

`PARTITION BY city`
`ORDER BY month`

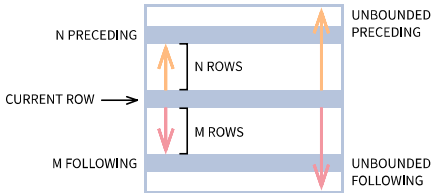
sold	city	month
300	Paris	1
500	Paris	2
200	Rome	1
300	Rome	2
400	Rome	3
100	London	1
400	London	2

Default ORDER BY: With no `ORDER BY` clause, the order of rows within each partition is arbitrary.

WINDOW FRAME

A **window frame** is a set of rows that are somehow related to the current row. The window frame is evaluated separately within each partition.

`<ROWS | RANGE | GROUPS> BETWEEN` *`lower_bound`* `AND` *`upper_bound`*



The bounds can be any of the five options:

- `UNBOUNDED PRECEDING`
- `n PRECEDING`
- `CURRENT ROW`
- `n FOLLOWING`
- `UNBOUNDED FOLLOWING`

The `lower_bound` must be BEFORE the `upper_bound`.

`ROWS` `BETWEEN 1 PRECEDING AND 1 FOLLOWING`

	city	sold	month
	Paris	300	1
	Rome	200	1
	Paris	500	2
	Rome	100	4
current row	Paris	200	4
	Paris	300	5
	Rome	200	5
	London	200	5
	London	100	6
	Rome	300	6

1 row before the current row and 1 row after the current row

`RANGE` `BETWEEN 1 PRECEDING AND 1 FOLLOWING`

	city	sold	month
	Paris	300	1
	Rome	200	1
	Paris	500	2
	Rome	100	4
current row	Paris	200	4
	Paris	300	5
	Rome	200	5
	London	200	5
	London	100	6
	Rome	300	6

values in the range between 3 and 5 `ORDER BY` must contain a single expression

`GROUPS` `BETWEEN 1 PRECEDING AND 1 FOLLOWING`

	city	sold	month
	Paris	300	1
	Rome	200	1
	Paris	500	2
	Rome	100	4
current row	Paris	200	4
	Paris	300	5
	Rome	200	5
	London	200	5
	London	100	6
	Rome	300	6

1 group before the current row and 1 group after the current row regardless of the value

As of 2024, GROUPS is only supported in PostgreSQL 11 and up.

ABBREVIATIONS

ABBREVIATION	MEANING
UNBOUNDED PRECEDING	BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
n PRECEDING	BETWEEN n PRECEDING AND CURRENT ROW
CURRENT ROW	BETWEEN CURRENT ROW AND CURRENT ROW
n FOLLOWING	BETWEEN CURRENT ROW AND n FOLLOWING
UNBOUNDED FOLLOWING	BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING

DEFAULT WINDOW FRAME

If `ORDER BY` is specified, then the frame is `RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW`.

Without `ORDER BY`, the frame specification is `ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING`.

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LIST OF WINDOW FUNCTIONS

Aggregate Functions
- `avg()`
- `count()`
- `max()`
- `min()`
- `sum()`

Ranking Functions
- `row_number()`
- `rank()`
- `dense_rank()`

Distribution Functions
- `percent_rank()`
- `cume_dist()`

Analytic Functions
- `lead()`
- `lag()`
- `ntile()`
- `first_value()`
- `last_value()`
- `nth_value()`

AGGREGATE FUNCTIONS

- **`avg(expr)`** – average value for rows within the window frame
- **`count(expr)`** – count of values for rows within the window frame
- **`max(expr)`** – maximum value within the window frame
- **`min(expr)`** – minimum value within the window frame
- **`sum(expr)`** – sum of values within the window frame

ORDER BY and Window Frame: Aggregate functions do not require an ORDER BY. They accept window frame definition (ROWS, RANGE, GROUPS).

RANKING FUNCTIONS

- **`row_number()`** – unique number for each row within partition, with different numbers for tied values
- **`rank()`** – ranking within partition, with gaps and same ranking for tied values
- **`dense_rank()`** – ranking within partition, with no gaps and same ranking for tied values

city	price	row_number	rank	dense_rank
		over(order by price)		
Paris	7	1	1	1
Rome	7	2	1	1
London	8.5	3	3	2
Berlin	8.5	4	3	2
Moscow	9	5	5	3
Madrid	10	6	6	4
Oslo	10	7	6	4

ORDER BY and Window Frame: `rank()` and `dense_rank()` require ORDER BY, but `row_number()` does not require ORDER BY. Ranking functions do not accept window frame definition (ROWS, RANGE, GROUPS).

DISTRIBUTION FUNCTIONS

- **`percent_rank()`** – the percentile ranking number of a row—a value in `[0, 1]` interval: (rank-1) / (total number of rows - 1)
- **`cume_dist()`** – the cumulative distribution of a value within a group of values, i.e., the number of rows with values less than or equal to the current row's value divided by the total number of rows; a value in `(0, 1]` interval

`percent_rank() OVER(ORDER BY sold)`

city	sold	percent_rank
Paris	100	0
Berlin	150	0.25
Rome	200	0.5 ★
Moscow	200	0.5
London	300	1

★ without this row 50% of values are less than this row's value

`cume_dist() OVER(ORDER BY sold)`

city	sold	cume_dist
Paris	100	0.2
Berlin	150	0.4
Rome	200	0.8 ★
Moscow	200	0.8
London	300	1

★ 80% of values are less than or equal to this one

ORDER BY and Window Frame: Distribution functions require ORDER BY. They do not accept window frame definition (ROWS, RANGE, GROUPS).

ANALYTIC FUNCTIONS

- **`lead(expr, offset, default)`** – the value for the row *offset* rows after the current; *offset* and *default* are optional; default values: *offset* = 1, *default* = NULL
- **`lag(expr, offset, default)`** – the value for the row *offset* rows before the current; *offset* and *default* are optional; default values: *offset* = 1, *default* = NULL

`lead(sold) OVER(ORDER BY month)`

order by month ↓

month	sold	lead
1	500	300
2	300	400
3	400	100
4	100	500
5	500	NULL

`lag(sold) OVER(ORDER BY month)`

order by month ↓

month	sold	lag
1	500	NULL
2	300	500
3	400	300
4	100	400
5	500	100

`lead(sold, 2, 0) OVER(ORDER BY month)`

order by month ↓ offset = 2

month	sold	lead
1	500	400
2	300	100
3	400	500
4	100	0
5	500	0

`lag(sold, 2, 0) OVER(ORDER BY month)`

order by month ↓ offset = 2

month	sold	lag
1	500	0
2	300	0
3	400	500
4	100	300
5	500	400

- **`ntile(n)`** – divide rows within a partition as equally as possible into *n* groups, and assign each row its group number.

`ntile(3)`

city	sold		ntile
Rome	100	1	1
Paris	100		1
London	200		1
Moscow	200	2	2
Berlin	200		2
Madrid	300		2
Oslo	300	3	3
Dublin	300		3

ORDER BY and Window Frame: `ntile()`, `lead()`, and `lag()` require an ORDER BY. They do not accept window frame definition (ROWS, RANGE, GROUPS).

- **`first_value(expr)`** – the value for the first row within the window frame
- **`last_value(expr)`** – the value for the last row within the window frame

`first_value(sold) OVER (PARTITION BY city ORDER BY month)`

city	month	sold	first_value
Paris	1	500	500
Paris	2	300	500
Paris	3	400	500
Rome	2	200	200
Rome	3	300	200
Rome	4	500	200

`last_value(sold) OVER (PARTITION BY city ORDER BY month RANGE BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING)`

city	month	sold	last_value
Paris	1	500	400
Paris	2	300	400
Paris	3	400	400
Rome	2	200	500
Rome	3	300	500
Rome	4	500	500

Note: You usually want to use `RANGE BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING` with `last_value()`. With the default window frame for `ORDER BY`, `RANGE UNBOUNDED PRECEDING`, `last_value()` returns the value for the current row.

- **`nth_value(expr, n)`** – the value for the *n*-th row within the window frame; *n* must be an integer

city	month	sold	nth_value
Paris	1	500	300
Paris	2	300	300
Paris	3	400	300
Rome	2	200	300
Rome	3	300	300
Rome	4	500	300
Rome	5	300	300
London	1	100	NULL

ORDER BY and Window Frame: `first_value()`, `last_value()`, and `nth_value()` do not require an ORDER BY. They accept window frame definition (ROWS, RANGE, GROUPS).

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